

Patricia García Caspueñas

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RESEARCH INTERESTS

- Data-driven engineering optimization, aerodynamic multi-fidelity data, Gaussian Processes, and active learning.
- Physics-informed Machine Learning techniques applied to experimental (PIV) and computational fluid dynamics.
- Reduced-order modeling (POD, AE, Galerkin), sparse system identification (SINDy), and data-driven forecasting (LSTM).

EDUCATION

UW | PHD IN MECHANICAL ENGINEERING

Seattle, WA, USA | 2025 - present

Advisors: Steven L. Brunton, Ana Larrañaga Janeiro

- Grade: 4.00/4.00 GPA.

UC3M | MSC IN AERONAUTICAL ENGINEERING (ENGLISH) [✉](#)

Madrid, Spain | 2022 - 2024

Advisor: Stefano Discetti

- Thesis: *Autoencoders for reduction and prediction of fluid flows*. DOI: [10.13140/RG.2.2.24559.80805](#) [✉](#).
- Class rank #1. Grade: 9.52/10 (4.08/4.33 GPA). Graduated with honors in 63/90 credits.

UC3M | BSc IN AEROSPACE ENGINEERING (ENGLISH) [✉](#)

Madrid, Spain | 2018 - 2022

Advisor: Stefano Discetti

- Thesis: *Nonlinear system identification applied to aerodynamic flows*. DOI: [10.13140/RG.2.2.11138.03522](#) [✉](#)
- Class rank #1. Grade: 9.27/10 (4.02/4.33 GPA). Graduated with honors in 129/240 credits.

CPM TERESA BERGANZA | INTERMEDIATE DEGREE OF MUSIC

Madrid, Spain | 2012 - 2018

Conservatorio Profesional de Música Teresa Berganza.

- Instrument: Piano.
- Performed at the National Auditorium of Madrid as part of a piano quintet.

EXPERIENCE

UW | GRADUATE RESEARCH ASSISTANT

Seattle, WA, USA | Sep 2025 - present

Researcher in the Data-Driven Dynamics and Control lab [✉](#), focusing on active learning strategies integrated with uncertainty-aware surrogate models to accelerate expensive aerodynamic optimization processes.

- Member of the NSF AI Institute for Dynamic Systems [✉](#) and the Boeing Advanced Research Collaboration (BARC) lab [✉](#).
- Delivered technical presentations about multi-objective optimization and engaged in discussions with Senior Technical Fellows at Boeing as part of the Foundational Research & AI for Engineering project led by Prof. Steven L. Brunton.
- Implemented Gaussian Process Regression and an active learning single-objective optimization process for quantum gate calibration in Julia in collaboration with the Scalable Quantum Research lab [✉](#) and Sandia National Laboratories [✉](#).

ALPINE F1 TEAM | AERODYNAMICS SCIENCE & TECHNOLOGY INTERN

Enstone, UK | Jan 2024 - Oct 2024

Focused on enhancing measurement techniques and contributing to conceptual aerodynamic development to improve overall race car performance.

- Developed data-driven tools in Python to improve the performance of existing wind tunnel measurement techniques.
- Contributed to in-house codebase used by 100+ engineers for Particle Image Velocimetry data-processing using Git.
- Analyzed image, pressure and force-based wind tunnel data, including sensitivity studies of main aero effects across runs.
- Participated in wind tunnel experiments, from test specification definition stages to data quality studies and post-processing.
- Led technical, team-wide meetings to ensure management-user alignment with development of data science tools.

Conducted research in the Experimental Aerodynamics and Propulsion lab [for](#) the ERC-NEXTFLOW [project](#), aimed at developing numerical methods for an enhanced, low-order compact description of flows from incomplete time-resolved measurements.

- Designed flow reconstruction algorithms from fast-probe measurements and non-time-resolved velocity snapshots.
- Applied basis-change methods (e.g. Proper Orthogonal Decomposition (POD), Extended POD, Dynamic Mode Decomposition (DMD), and Galerkin projections), and fast-probe optimization and prediction (QR pivoting, Kalman filter).
- Implemented sparse non-linear dynamical system identification (SINDy) on reduced-order models (ROMs).
- Post-processed numerical simulation and experimental data to create training datasets for data-driven surrogates.
- Participated in Particle Image Velocimetry (PIV) experiments to model flow behavior across bluff bodies.

HONORS & AWARDS

BARC Research Excellence Award , Boeing-UW	2026
Mechanical Engineering Fellowship Award (Fall term) (\$17,158.00) , UW	2025
Third Prize at the National Young Researchers Award in the Engineering category (€3,000.00) , Ministry of Science, Innovation and Universities of Spain	2025
Best MSc Thesis in Aeronautical Engineering (€1,000.00) , COIAE	2024
Highest GPA (major valedictorian), Aeronautical Engineering Class of 2024 , UC3M	2024
Highest GPA (major valedictorian), Aerospace Engineering Class of 2022 , UC3M	2022
Excellence Award (€2,000.00) , Board of Trustees of UC3M	2021
Four consecutive Excellence Awards (€6,300.00) , Comunidad de Madrid	2019 - 2022
UC3M Access Award (€500.00), UC3M Talent Program	2018

PUBLICATIONS

JOURNAL RESEARCH PAPERS

- [1] **García-Caspueñas, P., Larrañaga, A., & Brunton, S. L. (2026).** Preference-based Active Learning for Surrogate Multi-Objective Aerodynamic Optimization. *In preparation*.
- [2] Walton, C., **García-Caspueñas, P., Zacchei, F., Larrañaga, A., Brunton, S. L., & Mouradian, S. (2026).** Active Learning for Calibrating Entangling Gates via Surrogate-Based Optimization. *arXiv preprint*. DOI: [arXiv.2607.00284](#) [↗](#)
- [3] Solera-Rico, A., **García-Caspueñas, P., Sanmiguel Vila, C., & Discetti, S. (2026).** The balance between compactness and forecast accuracy of data-driven latent-space reduced-order models in controlled wake flows. *arXiv preprint*. DOI: [arXiv.2607.24569](#) [↗](#)
- [4] Li-Hu, Q. L., **García-Caspueñas, P., Ianiro, A., & Discetti, S. (2025).** Model-based time super-sampling of turbulent flow field sequences. *Physical Review Fluids*, 10(8), 084901. DOI: [10.1103/2lqd-g9mt](#) [↗](#).

CONFERENCE CONTRIBUTIONS

- [1] Solera-Rico, A., **García-Caspueñas, P., Sanmiguel Vila, C., & Discetti, S. (2025).** Modeling the dynamics of controlled wake flows with machine learning. *2nd European Fluid Dynamics Conference (EFDC2)*. 26-29 August 2025, Dublin, Ireland.
- [2] Li Hu, Q. L., **García-Caspueñas, P., Ianiro, A., & Discetti, S. (2024).** Galerkin model for time super-sampling of PIV measurements. *21st International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics*. DOI: [10.55037/lxaser.21st.223](#) [↗](#).
- [3] **García-Caspueñas, P., & Discetti, S. (2022).** System Identification For Enhanced Flow Description From Multi-Pulse PIV. *20th International Symposium on Applications of Laser Techniques to Fluid Mechanics*. DOI: [10.55037/lxaser.20th.194](#) [↗](#). [Speaker]

INVITED TALKS

1. Boeing Advanced Research Collaboration Annual Research Symposium . Active Learning in Engineering Optimization (Data-Driven Engineering Enabling Research & Development) - *Invited seminar*. 28th May 2026

GRANTS & SCHOLARSHIPS

Carlos Zapata Grant  , Fundación Humanismo y Ciencia (€7,200.00 for 9 months)

2023

Departmental Collaboration Scholarship , Universidad Carlos III de Madrid (€2,000.00 for 32 weeks)

2021 - 2022

CONTRIBUTING RESEARCHER ON FUNDED PROJECTS

1. **QSCOUT**  : Fast, model-free calibration of trapped-ion entangling gates. Quantum Scientific Computing Open User Testbed collaboration with Sandia National Laboratories. PI: Sara Mouradian. **Apr 2026 - Sep 2026**
2. **ERC-NEXTFLOW**  : Development of the next-generation flow diagnostics concept aimed at flow control. Project funded by a Starting Grant of the European Research Council. *Budget: €1,499,062.* PI: Stefano Discetti. **2021 - 2025**

MENTORING

Marissa Till , Undergraduate student at Caltech, SURF  program (w/ S. L. Brunton & A. Larrañaga).

Mar 2026 - present

Project: *Multi-Objective Optimization for Sustainable Aviation Fuels.*

Yixuan Lin , Undergraduate student at Caltech, SURF  program (w/ S. L. Brunton & A. Larrañaga).

Jun 2026 - present

Project: *A Python version of the Immersed Boundary Projection Method for 2D airfoils.*

Aritz Schube , Undergraduate student at Caltech, SURF  program (w/ S. L. Brunton & A. Larrañaga).

Jun 2026 - present

Project: *Varying-fidelity flow solvers for aerodynamic shape optimization.*

SCIENTIFIC OUTREACH

Speaker in Madrid's XII Science Fair , *Desvelando los secretos de la fluidodinámica*
(Unveiling the secrets of fluid dynamics)

Mar 2023

Speaker in Madrid's Science and Innovation Week , *¡Aprende a volar con aviones de papel!*
(Learn to fly with paper airplanes!)

Nov 2022

ACADEMIC SERVICE

Peer Reviewer , *AIAA Journal*

2026 - present

Verified peer review record on Web of Science: 2 reviews completed 

SKILLS

MATLAB	Advanced	Python	Advanced
L ^A T _E X	Intermediate	MS Office	Intermediate
Git	Intermediate	Julia	Basic

LANGUAGES

Spanish	Native
English	C2 on the CEFR (155/160)
	Duolingo English Test
	ID: wl9tokeehbk7i9s5

PROFESSIONAL DEVELOPMENT

The Git & Github Bootcamp , Udemy

Jan 2025

Artificial Intelligence Concepts - Introduction to Machine Learning (Online) , *University of Oxford*

Jul 2024

Laser Hazard Awareness Training Course , *Lasermet*

Mar 2024

Artificial Intelligence and Deep Learning from scratch in Python , Udemy

Aug 2023